

ADVANCED DATA SCIENCE PROJECT

DressMe — Personal fashion assistant

1. Project overview

DressMe is an intelligent personal fashion assistant designed to transform a user's wardrobe into a smart and personalized fashion ecosystem.

Users can build a digital version of their wardrobe by uploading images of their clothing. Using AI, computer vision, and recommendation systems, the platform analyzes each item to understand its characteristics, including category, color, style, season, and compatibility with other clothes.

Based on the user's wardrobe and personal style, **DressMe** provides intelligent and personalized recommendations. It can suggest complete outfits, identify missing items, recommend products to complement an existing look, and help users make smarter purchasing decisions.

For example, before buying a new item, a user can ask: **"Should I buy this?"**

DressMe can analyze the item and determine whether it:

- matches the user's personal style
- can be combined with existing clothes
- is too similar to items the user already owns

The platform may also include a **virtual try-on** feature, allowing users to visualize selected outfits.

2. Your challenge

Design and develop an intelligent AI-powered platform that acts as a **personal fashion shopper**.

Your system should be able to understand the user's wardrobe and transform clothing data into meaningful and personalized fashion recommendations.

The challenge is to develop an intelligent solution capable of:

- Building a **digital wardrobe** from clothing images.
- Automatically analyzing and classifying clothing items.
- Understanding the user's personal style and preferences.

- Recommending outfits using clothes already available in the wardrobe.
- Suggesting items that can complete an existing outfit.
- Identifying gaps and missing pieces in the user's wardrobe.
- Detecting similar or redundant clothing items.
- Evaluating whether a new clothing item is a **good purchase**.
- Explaining why a recommendation or purchase is relevant.
- Optionally integrating a **Virtual Try-On** feature.

The solution should be designed with the **Tunisian market** in mind, considering local fashion styles, clothing categories, seasons, budgets, and shopping habits.

3. Expected outcomes

By the end of the project, you should deliver a functional prototype of an intelligent personal fashion assistant.

The expected solution should allow a user to:

- **Manage a smart wardrobe:** Upload and organize clothing items into a digital wardrobe.
- **Understand personal style:** Generate a personalized fashion profile based on the user's clothing, colors, styles, and preferences.
- **Complete an outfit:** Recommend missing items such as shoes, bags, jackets, or accessories to complete a look.
- **Make smarter purchases:** Analyze a potential purchase and provide recommendations such as:
 - ✓ **Recommended purchase:** *"This item matches your style and can be combined with several pieces in your wardrobe."*
 - ✓ **Think twice:** *"You already own similar items, and this purchase adds limited value to your wardrobe."*
 - ✓ **Not recommended:** *"This item has low compatibility with your current wardrobe and personal style."*
- **Visualize clothes:** Integrate virtual try-on to visualize clothing or complete outfits.

The final prototype should demonstrate the integration of relevant Data Science and AI techniques, such as: **Computer vision • machine learning • deep learning • recommendation systems • multimodal AI • generative AI • NLP**

4. Think beyond automation

You are encouraged to go beyond basic automation and explore innovative ideas such:

- Explainable AI for fashion recommendations
- Personalized compatibility scores
- AI-generated shopping advice
- Conversational fashion assistants
- Multimodal search using text and images
- Sustainable fashion recommendations
- Wardrobe optimization
- Budget-aware recommendations
- Context-aware recommendations based on occasion, season, or weather
- AI-powered virtual try-on

5. Livrables

Specifications, functional prototype application, final report, oral presentation.